

A group of children and adults are wading in shallow ocean water. In the center, a woman wearing a red cap and a white tank top is looking down at something in her hands. To her right, a man in a dark blue shirt is also looking down. Several children are scattered around them, some looking at the water and others looking towards the camera. The water is shallow and clear, revealing the sandy bottom. In the background, there is a concrete walkway and some greenery.

— A PILOT PROPOSAL FOR JOURNEY SCHOOL · TAMARINDO · 2026

Ocean Literacy at Journey School

A pilot program for school-wide marine science education.

— EFFI BAR · MARINE BIOLOGIST & EDUCATOR

Why we're here

- i.* Journey's mission speaks to **connection to the natural world** and meaningful experiences. That language is what brought me to this conversation.
- ii.* Guanacaste students live closer to the Pacific than almost any school I've worked with — and yet rarely engage with it scientifically.
- iii.* Today, before I show you anything, I want to listen first.

“

Before I show you anything — how do you currently approach marine and environmental education? What's working? What feels missing?

— THE QUESTION I'D LIKE TO START WITH

Why this fits Journey

a.

Inside the curriculum, not bolted on.

The Ocean Literacy framework (NOAA / NMEA) maps to IB PYP *Sharing the Planet* and to MEP biodiversity standards. We integrate; we don't add a subject.

STANDARDS-ALIGNED

b.

A real draw for parents.

A visible admissions differentiator with a credential pathway — FEE Green Flag, plus evidence that strengthens MSA and Cognia portfolios.

ADMISSIONS & ACCREDITATION

c.

Career exposure from middle school.

Most students don't know marine careers exist — boat captain, researcher, conservation officer, marine policy. The program lets them picture themselves in those futures.

FUTURE PATHWAYS

What is ocean literacy?

An internationally recognized framework — not my invention.

- i.* Developed by **NOAA and the National Marine Educators Association**; in active use in 100+ countries.
- ii.* Seven core principles, 45 supporting concepts, with a full K–12 scope and sequence.
- iii.* Maps directly onto IB PYP *Sharing the Planet* and MEP *Ciencias Naturales* — *biodiversidad marina*.

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MEP mandates that biodiversidad marina be taught — but most schools fulfill it with terrestrial content because they don't have a marine biologist. I fill that gap.

— THE GAP I FILL

The seven principles

01 Earth has one big ocean with many features.

03 The ocean is a major influence on weather and climate.

05 The ocean supports a great diversity of life and ecosystems.

07 The ocean is largely unexplored.

02 The ocean and life in the ocean shape the features of Earth.

04 The ocean made Earth habitable.

06 The ocean and humans are inextricably interconnected.

How it scales by grade

K–2
PYP EARLY

"The ocean is alive, and I've met it."

Storytelling, touch tanks, habitat models, tide-pool encounters.

3–5
PYP LATE

"The ocean is a system, and I can describe it."

Ecosystems, beach-debris counts, water-clarity logs, vocabulary.

6–8
MYP

"The ocean has problems, and I can investigate them."

Water-quality testing, citizen science, ecosystem dynamics.

9–12
DP / CAS

"The ocean has answers, and I can find them."

Research-question fieldwork, marine policy, NOAA / PRETOMA datasets.

What students do



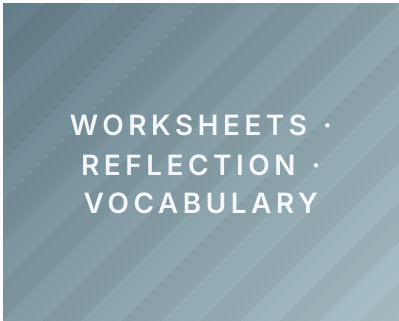
Hands-on 90-minute lessons
Rocky-shore models, plankton towing, lab work.
Inquiry-driven, materials provided.



Field excursions
Avellanas and Negra tide pools, sea turtle nesting
beaches, mangroves, Gulf of Nicoya.



Real data collection
Water clarity, beach debris, microplastics — feeding
citizen science platforms.



Structured worksheets & discussion
Rigor, not just memory: pre-/post-fieldwork prompts,
vocabulary, reflection.

The local advantage



OSTIONAL · 45 MIN

Olive ridley arribadas.

August – November · school year

LAS BAULAS · 15 MIN

Leatherbacks, critically endangered.

October – March · school year

TAMARINDO · HERE

Estuary, mangroves, year-round access.

Every term

No PowerPoint beats — "the turtles are nesting 20 minutes from here, and your Grade 7 class can document it."

Two ways in

OPTION 1 · RECOMMENDED

Full year

- Begins December, runs the academic year
- 9 × 90-minute classroom lessons per grade band
- Monthly visits with teacher co-planning
- 3 field excursions (tide pools · turtles · mangroves)
- 2–3 teacher PD sessions, full materials provided
- Assessment artefacts for MSA & Cognia portfolios

Designed for renewability — teacher capacity outlasts the contract.

OPTION 2 · LIGHTER TOUCH

Single project

- One ocean literacy project per grade level
- 1 field excursion
- Project-based assessment
- 2 teacher trainings
- A clean way to test the fit before committing

About Effi



Effi (Efrat) Bar — marine biologist & environmental educator, 10+ years.

TAMARINDO · COSTA RICA

- B.Sc. Marine Sciences (with honors); graduate work in marine protected area management.
- EcoOcean ecological surveys and reserve planning.
- Educational programs in classroom and field, including aboard research vessels.
- Public outreach with sea turtle rescue and rehabilitation.

“*I live here. My family is rooted here. The program is designed to be renewable — I build teacher capacity so the school keeps the framework regardless of me.*”

What I'd love to leave with today

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- 01 The **one problem** in marine or environmental education you can't solve with what you have today.
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- 02 The **age band(s)** you'd prioritize for the pilot.
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- 03 The **shape** that fits your calendar — weekly · special days · camps · transversal.
-
- 04 Whether you'd like a **tailored written proposal** this week.

“

I want it to be something built for Journey specifically — not a template. That's why I'm asking these questions before I write it, not after.



Thank you.

Questions?

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